PEARLCARE MANAGEMENT SYSTEM

CASE STUDY: KAMPALA SKIN CLINIC

A PROJECT REPORT

Submitted

In partial fulfilment of the requirement for the award of

DIPLOMA

IN

SOFTWARE ENGINEERING

(DSE)

By

LWEGABA MICKPAUL KUTEESA

Roll No. 022240760

Under the guidance
of
Mr. TOBIAS KAKOOZA

(Supervisor)

FACULTY OF INFORMATION & COMMUNICATION TECHNOLOGY

DEPARTMENT OF DIPLOMA STUDIES

ISBAT University-City Campus

P.O Box 8383, Conrad Plaza, Kampala

JUNE 2026

DECLARATION

I, **LWEGABA MICKPAUL KUTEESA**, bearing registration number **022240760**, hereby solemnly declare that this Project Report, titled **PEARLCARE MANAGEMENT SYSTEM** is a meticulously crafted and entirely original Final Year Project work. It has been executed under my intellectual purview, with due diligence and academic integrity.

I affirm that this document has not been previously submitted for evaluation or recognition to any educational institution or for any other award.

The ideas, concepts, and findings presented herein are the result of my independent research efforts. I take full responsibility for the content and authenticity of this report, and I am committed to upholding the highest standards of academic honesty.

Date: ……………………………………………….

Signed: …………………………………………....

DEDICATION

This work is dedicated to my uncle, Mr. SSEMPIJJA MIKE whose unwavering financial support made my education possible. Your investment in me was not just monetary, it was a belief that I could build something meaningful, and this system stands as proof that your sacrifice was not in vain.

To my parents, Ms. BERNADETTE NDAGIRE and Mr. PAUL MUGENYI, the reason persistence is not optional for me. You carried burdens I didn't fully understand at the time, yet you never let me feel their weight. This project stands on the strength you showed when it would have been easier to break.

And to Dr. Fred Kambugu, whose skill and decisive care turned a critical moment into a second chance. What you did was not theoretical, it was real, it was timely, and it mattered. This system is built with that same standard in mind, that efficiency and precision in healthcare are not luxuries, but necessities.

This is more than a project. It is a response to what I have been given and a commitment not to waste it.

DEPARTMENT OF DIPLOMA STUDIES

BONAFIDE CERTIFICATE

This is to certify that the project report titled **PEARLCARE MANAGEMENT SYSTEM** is the bonafide work of **LWEGABA MICKPAUL KUTEESA**, who has carried out the project under the guidance and supervision of the **ASSISTANT ACADEMIC REGISTRAR** and the **HEAD OF DEPARTMENT.**

The project work was completed as part of the requirements for the Diploma program in Software Engineering and is presented in fulfillment of the academic requirements of ISBAT University.

We hereby affirm that this project has been conducted with due diligence, adhering to the academic standards and ethical guidelines established by the institution.

ASSISTANT ACADEMIC REGISTRAR	**HEAD OF DEPARTMENT**
SIGNATURE WITH DATE	SIGNATURE WITH DATE

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APPROVAL

I hereby certify that I have reviewed and approved the content of this final year book project. The work presented in this book represents the student's original ideas and research, and demonstrates their knowledge and understanding of the subject matter. I believe this project reflects the student's dedication and hard work, and I am confident that it meets the academic standards of excellence expected at this level. The work is now ready for submission to the Academic Committee of ISBAT University with my approval.

Signature: ………………………………………

MR.TOBIAS KAKOOZA

(SUPERVISOR)

Date: ……………………………………………………

ACKNOWLEDGEMENT

I express my profound gratitude to the Almighty God for granting me the wisdom, strength, and perseverance required to successfully complete this work. His guidance has been instrumental throughout this academic journey.

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CPU	Central Processing Unit
CSS	Cascading Style Sheet
DFD	Data Flow Diagram
ER	Entity Relationship
PHP	Hypertext Preprocessor
RAM	Random Access Memory
ROM	Read Only Memory
SQL	Structured Query Language

ABSTRACT

Healthcare facilities in Uganda often rely on manual or semi-digital systems for patient queuing, record management, and operational monitoring. These approaches lead to prolonged waiting times, inefficient patient flow, data fragmentation, and limited visibility into clinic performance. This project presents the design and implementation of an intelligent clinic queue, patient management, and operational analytics system tailored to the Ugandan clinical workflow.

The system integrates automated queue management with real-time patient registration, consultation tracking, and centralized electronic medical records. It introduces a structured, priority-based queuing mechanism that optimizes patient flow while accommodating emergency and special cases. Additionally, the system incorporates an analytics module that provides actionable insights into clinic operations, including patient volumes, service times, and staff performance.

Built using modern web-based technologies, the system ensures accessibility, scalability, and ease of integration into existing clinic environments. By digitizing and streamlining core processes, the proposed solution reduces patient waiting times, improves record accuracy, and enhances decision-making through data-driven insights.

The implementation demonstrates how context-aware digital systems can significantly improve efficiency and service delivery in resource-constrained healthcare settings such as Uganda.

CHAPTER 1
INTRODUCTION

1.0 Introduction

The Uganda Vision 2040 identified Universal Health Coverage as a key pillar in the fight against poverty. Achieving this vision requires increasing access to quality health services for all Ugandans, with a focus on specialized care that has historically been limited in availability. Dermatological conditions represent a significant yet underserved area of healthcare in Uganda, affecting individuals across all demographics and geographic regions.

The Ministry of Health Strategic Plan 2021/2025 emphasized strategies to expand health service delivery and improve health information management across all specialties. In the field of skin health, clinics such as Kampala Skin Clinic play a vital role in providing specialist dermatological services. Effective management of patient data, appointment scheduling, and treatment records is essential to delivering high-quality dermatological care.

Joining in the effort to improve health information management in Uganda, this project aims to develop a digital system to reliably collect and manage patient data for Kampala Skin Clinic. The system is designed to enable secure storage, streamlined appointment management, and coordinated sharing of clinical data without violating the patient's privacy. The main aim of the project is to put the owner of the health data, the patient, in control of their information while empowering clinicians to deliver efficient, evidence-based dermatological care.

This chapter introduces the project and the environment in which it was developed. It examines the state of health information systems in Uganda with a focus on dermatological services. Thereafter, it identifies the problem facing skin clinic management, data collection, and patient record keeping before proposing a solution. It establishes the main and specific objectives of the project, the scope, and the significance of the study. The chapter closes with the limitations of this project.

1.1 Background of Study

This project was conducted at Kampala Skin Clinic in Uganda. Kampala Skin Clinic is a private specialist dermatology clinic located in Kampala, the capital city of Uganda. The clinic is situated in the Kampala Central Division and serves patients from across Kampala and the wider country who require specialist skin, hair, and nail care. It is conveniently located along a major road in the city, making it accessible to a wide patient population.

The clinic was established to address the growing demand for professional dermatological care in Uganda, where skin conditions such as eczema, psoriasis, acne, fungal infections, and skin cancer represent a significant burden on the population. Since its founding, Kampala Skin Clinic has grown steadily to become one of the leading skin care facilities in the country.

The clinic has expanded its services over time in response to increasing patient numbers and the diversification of dermatological conditions presenting for treatment. It now offers comprehensive dermatological services including diagnosis, medical and cosmetic treatment, and follow-up care for a broad range of skin conditions.

The facility offers a range of specialized services across different departments and clinics including but not limited to:

- **Medical Dermatology:** diagnosis and treatment of skin, hair, and nail disorders including acne, eczema, psoriasis, vitiligo, fungal infections, and dermatitis.
- **Cosmetic and Aesthetic Services:** including chemical peels, laser therapy, botox, skin brightening treatments, and scar management.
- **Surgical Dermatology:** handling excision of skin lesions, moles, cysts, skin tags, and biopsy procedures.
- **Paediatric Dermatology:** specializing in skin conditions affecting children including nappy rash, childhood eczema, and birthmarks.
- **Trichology Services:** dealing with hair loss conditions such as alopecia areata, androgenic alopecia, and scalp disorders.
- **Diagnostic and Laboratory Services:** including skin patch testing, allergy testing, dermoscopy, and histopathology.

At the time of this project (2025), the clinic was served by a team of qualified dermatologists, clinical officers, nurses, and support staff, attending to a significant number of patients weekly from across Kampala and the surrounding regions.

1.2 Literature Review

The advent of computer technology has led to the widespread use of information systems across all industries. Information Systems are integrated sets of components that collect, store, process, and disseminate information to support decision-making and organizational operations. The importance of information systems in healthcare cannot be overstated. In the health industry particularly, information systems play a vital role in decision making, planning, analytics, monitoring, and in several other ways.

In recent years, Uganda has seen the development and deployment of several health information systems. From health data systems to monitoring and evaluation systems, the health industry in Uganda is using information systems to ensure delivery of high-quality health services and extending several health services to all Ugandans. These systems have positively affected the quality of services rendered to Ugandans through better planning, improved supervision, and better service delivery. They have also made available data for planning and research.

In the field of dermatology, specialized clinics such as Kampala Skin Clinic face unique challenges in managing patient health data. Skin conditions often require long-term follow-up, photographic documentation of lesion progression, tracking of treatment responses, and careful management of prescriptions for topical and systemic therapies. With the increased use of information systems, new challenges have arisen including data collection inaccuracies, lack of centralized records, and difficulty in continuity of care when patients move between providers.

The decentralized nature of current systems presents a challenge where patient skin health information needs to be reliably shared and made available across consultations. Human health information should be stored securely in a world riddled with threats to confidentiality and privacy. It is therefore in regard to these challenges that this project, PEARLCARE MANAGEMENT SYSTEM, was deemed necessary and relevant to the state of dermatological care and health information systems at Kampala Skin Clinic in Uganda.

1.3 Problem Statement

Owing to the importance of health data in the delivery of dermatological services and clinical decision making, it is essential that patient health data is handled securely and reliably. Moreover, the data must be managed by a system that makes it readily available to facilitate patient appointments, treatment tracking, prescription management, and health conditions monitoring. The absence of a dependable clinic management system that makes this possible for individuals receiving dermatological care services at Kampala Skin Clinic is the motivation for this project.

1.4 Main Objective

To develop a centralized skin clinic management system for Kampala Skin Clinic, which gives patients and health workers control over clinical data with no hindrance to service delivery, by allowing dermatologists and clinical staff secure and lawful access to patient health data when the need arises.

1.5 Specific Objectives

- To encourage patients at Kampala Skin Clinic to participate in the collection and analysis of their dermatological health data by tracking patient skin health information beyond the confines of the clinic.
- To make a more complete set of clinical records for each patient readily available to dermatologists and health workers at Kampala Skin Clinic by eliminating errors of omission that occur when patients rely on memory and inefficient paper-based record keeping mechanisms.
- To make patient skin health information sharable between patients and dermatologists at Kampala Skin Clinic at the patients' discretion.

1.6 Scope of Study

This project focuses on 5 key areas of dermatological health data generation and management at Kampala Skin Clinic:

1. Patient skin health data collection
2. Appointment and treatment timeline management
3. Patient health data protection
4. Secure data sharing
5. Integration with existing health systems

1.6.1 Patient Skin Health Data Collection

The project aims to empower patients and dermatologists at Kampala Skin Clinic to take part in collecting and managing patient skin health data by providing a platform through which they can track symptoms, skin condition progression, photographic records, and provide secure access to emergency health data when needed.

1.6.2 Appointment and Treatment Timeline Management

The project aims to improve how patients manage their dermatological therapies, specifically the administration of prescriptions and scheduled treatments, by providing appointment scheduling, reminders, and alerts to users when therapies and follow-up consultations need to be attended.

1.6.3 Patient Health Data Protection

Security of health data collected from patients is explored in this project by focusing on the three pillars that make up good security in any system: *Availability* is ensured by making data present when needed and at the lowest possible cost; *Confidentiality* is emphasized by implementing role-based access control and granting the owner of the data, the patient, absolute control over their data meaning rights to grant access or deny access to their data. *Integrity* is ensured by ensuring that data collected can be valid and can be trusted, and that data in storage and in transit is not compromised and access to the data is only by authorized parties.

1.6.4. Secure Data Sharing

During data sharing, measures are taken to ensure that the integrity of the data is not compromised. Tactics like encryption via SSL/TLS and transmission over HTTPS are explored to safeguard all patient dermatological records during transmission.

1.6.5 Integration with Existing Systems

Finally, the project explores ways to share the data collected by the proposed system with stakeholders who use other platforms but need to access collected patient skin health data. It goes further to leverage existing systems to perform actions like verifying identity and performing authentication among other things.

1.7 Significance of the Study

The project, once completed successfully, will see more involvement of patients in their skin health data collection and analysis. Dermatologists and health workers at Kampala Skin Clinic shall be rewarded with more complete data for analysis, diagnosis, and treatment planning. Other stakeholders, like the Ministry of Health and national health authorities, will have available public skin health data for planning, decision making, and forecasting of public health scenarios related to dermatological conditions in Uganda.

In the short run, this shall be realized by encouraging patients and health workers to download and utilize the application. Users of the application shall be taught to capture relevant and correct information. Dermatologists and clinical staff shall be encouraged to take part in encouraging patient involvement, and to use the collected skin health data appropriately. Integrating with HMIS and other public health information systems will provide data for other stakeholders.

1.8 Limitations of the Study

This project, PEARLCARE MANAGEMENT SYSTEM, while revolutionary to the health information systems space and dermatological care in general, is not without limitations. A few of the limitations that were identified are outlined below.

1.8.1 Technical Limitation

The implementation and utilization of this project is affected by the availability of technical infrastructure in areas where users reside. It is also affected by knowledge of and ability to interact with specific technologies. It is therefore evident that the project will focus on urban centres with better-developed infrastructure and technical knowledge. Eventually, with the rapid technological development in Uganda and Kampala in particular, the project should expand to serve users across the country.

1.8.2 Security Limitation

The security risk posed by handling patient skin health data in a distributed system, while mitigated by implementation of access control, validation, and privacy policies, still remains. Data is collected at the discretion of the user (patient) which still poses a risk of false data and omissions. This shall be overcome with continuous training and sensitization of users.

1.8.3 Social-Economic Limitation

The project is not oblivious to the bias which is associated with health data collected and provided by patients without direct clinical supervision at all times. In this light, the project faces the barrier of educating all parties involved to improve collaboration and increase trust, while maintaining quality of clinical information and service delivery.

It is also evident that transferring rights of dermatological health data management from the clinic to the individuals from whom data is collected might pose a financial risk to facilities that rely on holding data to encourage patients to keep returning. With this in mind, there is expected to be some opposition to adaptation of the system by such facilities. This is a temporary limitation however. After the project is deployed and widely adopted, convincing the reluctant parties of the benefit of entrusting skin health data to the patients will become considerably easier.

CHAPTER 2
SYSTEM ANALYSIS AND REQUIREMENTS

2.1 Introduction

Before any new system is built, it is important to understand what is already in place and why it is not working well enough. This chapter takes a close look at how Kampala Skin Clinic currently handles patient records and clinic operations, and then builds the case for a better solution. It covers a feasibility study to check whether the proposed system is practical, looks at how patient data will be managed, and lists the specific requirements the system must meet in terms of software, hardware, and the environment it will run in.

The goal of this chapter is to make sure that the system being built is the right fit for Kampala Skin Clinic, its staff, and its patients. Every design decision made later in this project is grounded in the findings presented here.

2.2 Existing System

At the time this project was started, Kampala Skin Clinic was managing its patient records and day-to-day operations mostly through a combination of paper files and basic digital tools like spreadsheets and shared folders. While these methods have served the clinic for a number of years, they have clear gaps that affect both the quality of service and the experience of patients.

How the Current System Works

When a new patient visits the clinic, their personal and medical details are written into a physical patient card or file. This file is stored in a cabinet at the reception desk and retrieved each time the patient comes back. For returning patients, staff must search through the files manually, which takes time, especially when the clinic is busy.

Appointment scheduling is done verbally or through phone calls, and appointments are sometimes recorded in a diary or basic spreadsheet. Prescriptions and treatment notes are written by hand on paper and stored in the patient file. Follow-up instructions are given verbally to the patient, meaning there is a high chance that patients forget what they were told.

Some staff members use their personal phones or email to share patient information with colleagues, which raises serious concerns about patient confidentiality. There is no central place where all patient data is stored and accessed securely.

Problems with the Current System

The current system has a number of well-known problems that affect both the clinic and its patients:

- **Lost or damaged records:** Paper files can be misplaced, torn, or damaged by water or fire. Once a record is lost, there is no backup and the patient history is gone.
- **Slow service:** Looking for patient files manually slows down the process at reception, leading to long waiting times for patients.
- **No appointment reminders:** Patients are not reminded of their next visit, which leads to many missed follow-up appointments and poor treatment outcomes, especially for conditions like eczema or psoriasis that require ongoing management.
- **Poor treatment tracking:** It is hard to track how a patient's skin condition has changed over time because there is no structured way to compare past and current records. There are no photos or visual records attached to the file.
- **Unsafe data sharing:** When staff share patient data over personal phones or email, there is a risk of that information falling into the wrong hands.
- **Limited reporting:** The clinic cannot easily generate reports on things like the most common conditions being treated, patient return rates, or medicine usage because the data is spread across many unorganised files.
- **No patient access to their own records:** Patients have no way to view their own medical history, past prescriptions, or treatment notes. They depend entirely on clinic staff to retrieve and share that information.

These problems make it clear that Kampala Skin Clinic needs a better system, one that is digital, organized, secure, and easy to use for both staff and patients.

2.3 Feasibility Study

A feasibility study was done to check whether it is actually possible to build and use the proposed PearlCare Management System. Three key areas were looked at: technical feasibility, economic feasibility, and operational feasibility.

Technical Feasibility

The system is being built using tools and technologies that are already widely available and well understood. The frontend will be built using Vanilla JS, which allows the same application to run smoothly on the web. The backend will use PHP with a MySQL database, both of which are free to use and well supported by a large community of developers.

Kampala Skin Clinic already has internet access and at least basic smartphones are used by both staff and patients. The system will be hosted on a cloud server, which means it does not require the clinic to buy expensive hardware. As long as there is an internet connection, the system can be accessed from any device. From a technical standpoint, building and running this system is entirely possible.

Economic Feasibility

The cost of building the system is manageable. Since all the development tools being used are open source and free, the main costs involved are development time and cloud hosting fees. Cloud hosting services like AWS or DigitalOcean offer affordable plans that suit the size of a clinic like Kampala Skin Clinic.

The long-term savings from switching to a digital system are significant. The clinic will spend less on paper, printing, and physical file storage. Staff will spend less time on administrative tasks, which means more time can be given to patient care. Fewer missed appointments also mean more consultations completed, which directly improves the clinic's income. Overall, the system is economically worth building and running.

Operational Feasibility

The system has been designed with ease of use in mind. The interface is clean and simple, so staff do not need to be highly technical to use it. Patients interact with the system through the web where they can view their records, book appointments, and receive reminders.

A short training session for clinic staff will be enough to get them up and running. Patients will be guided through the application during their first visit to the clinic. Because the system is solving very real and frustrating problems that staff and patients already deal with every day, there is strong motivation for people to adopt it. Operationally, the system is very feasible.

2.4 Information and Database Management

One of the most important parts of this system is how patient data is stored and managed. Skin conditions often require long-term care, which means a patient might be visiting the clinic for months or even years. During that time, a lot of data is collected, including personal details, skin condition history, photos, prescriptions, appointment records, and treatment notes. All of this needs to be organized in a way that is easy to retrieve, safe to store, and accurate.

Database Design

The system uses a relational database (MySQL) to store all data. In a relational database, information is stored in tables, and the tables are linked to each other. For example, the patient table is linked to the appointments table, which is linked to the consultations table, which is linked to the prescriptions table. This means that with just a patient's ID, the system can pull up their full history in seconds.

The main tables in the database include:

- **Patients:** stores personal details like name, date of birth, contact information, and residence.
- **Consultations:** stores notes made by the dermatologist during each visit, including the diagnosis and treatment plan.
- **Prescriptions:** stores medicines prescribed, dosage, frequency, and duration.
- **Skin Condition Records:** stores details about each condition a patient has been diagnosed with, including photos and progress notes over time.
- **Users:** stores login credentials and roles for staff and patients.

Data Security and Privacy

Patient health data is personal and sensitive. The system takes several steps to protect it. All data is encrypted before it is stored in the database, and all communication between the app and the server is protected using HTTPS and SSL/TLS encryption. This means that even if someone managed to intercept the data in transit, they would not be able to read it.

Access to the system is controlled by roles. A receptionist can book appointments and view basic patient details, but cannot change a doctor's notes. A dermatologist can view and edit clinical records for their patients, but cannot access the admin settings. Patients can only see their own records. This role-based approach ensures that each person only sees what they are supposed to see.

Regular backups are made automatically to make sure that data is not lost in case of a technical failure. The backup copies are stored securely in a separate location from the main server.

Data Integrity

The system includes validation checks to make sure that only correct and complete data is entered. For example, a date of birth cannot be set in the future, a phone number must have the right number of digits, and a prescription cannot be saved without a diagnosis attached to it. These small checks go a long way in preventing errors that could affect patient care.

2.5 Requirement Specification

This section lists what the system must be able to do and what it needs to run properly. Requirements are split into functional requirements (what the system does) and non-functional requirements (how well it does it), as well as the specific software and hardware needed.

Functional Requirements

These are the key things the system must be able to do:

- Allow clinic staff to search for and view any patient record quickly.
- Allow patients and staff to book, reschedule, and cancel while managing the queue.
- Allow dermatologists to record consultation notes, diagnoses, and treatment plans.
- Allow dermatologists to issue and manage prescriptions digitally.
- Allow photos of skin conditions to be uploaded and linked to a patient's record.
- Track the progress of a patient's skin condition over multiple visits.
- Allow patients to share their records with another health provider of their choice.
- Generate basic reports on appointments, common conditions, and prescription usage.
- Support role-based login so that staff and patients each see only what is relevant to them.

Non-Functional Requirements

Beyond what the system does, these are the standards it must meet:

- **Speed:** the system should load patient records and complete common tasks in under three seconds.
- **Availability:** the system should be accessible at least 99% of the time during clinic working hours.
- **Security:** all patient data must be encrypted, and login must be protected with strong passwords and optionally two-factor authentication.
- **Usability:** the interface must be simple enough for a receptionist with basic computer skills to use after a short training session.
- **Scalability:** the system should be able to handle a growing number of patients and appointments without slowing down.

2.5.1 Software Requirements

The following software tools and platforms are required to build and run the Kampala Skin Clinic Management System:

Component	**Tool / Platform**	**Purpose**
Frontend (Web)	HTML5, CSS3, JS, AJAX	Web Application
Backend / API	PHP	Handles business logic and communication between app and database
Database	MySQL	Stores all patient, appointment, and clinical data
Authentication	JWT (JSON Web Tokens)	Manages secure login sessions for staff and patients
Cloud Hosting	Infinityfree	Hosts the backend server and database in the cloud
Version Control	Git / GitHub	Tracks code changes and supports team collaboration
Operating System	Windows(Development)	Development and production environments

Figure 1: Software Requirements..

All the software tools listed above are either open source or have free tiers that are sufficient for the scale of this project. This keeps the overall cost of development low while still using industry-standard tools.

2.5.2 Hardware Requirements

The hardware requirements for this system are modest because most of the processing happens on the cloud server rather than on the user's device. Below are the minimum hardware requirements for each part of the system:

For the Cloud Server

- At least 2 CPU cores
- At least 4 GB of RAM
- At least 50 GB of storage (SSD preferred for faster database access)
- Stable internet connection with at least 10 Mbps upload speed

For Clinic Staff (Receptionist, Nurses, Records and Doctors)

- A smartphone or computer with a modern web browser.
- At least 2 GB of RAM on the device.
- Internet connection (Wi-Fi or mobile data)
- A camera or connected webcam for uploading skin condition photos (optional but recommended)

These requirements are realistic for Kampala, where smartphone use is common and internet connectivity through mobile data networks is widely available. The system does not require any special or expensive hardware, which makes it practical for a clinic like Kampala Skin Clinic to adopt.

2.6 Operational Environment

The system will operate in a real clinical setting at Kampala Skin Clinic, a busy dermatology clinic in Kampala, Uganda. Understanding this environment is important because it shapes many of the design decisions made throughout the project.

Clinic Environment

Kampala Skin Clinic operates Monday to Saturday and sees a large number of patients each week. The reception area is the main point of contact for patients, and the system must work reliably in this fast-paced environment. Receptionists need to quickly check in patients, and pull up records without any delays.

Consultation rooms are where dermatologists and clinical officers interact with patients. They need to access patient histories, record new notes, and issue prescriptions during or after each consultation. The system must be easy to use in this setting, where the focus should be on the patient, not on navigating a complicated interface.

Connectivity

Kampala has generally good mobile data coverage, and the clinic has a stable Wi-Fi connection. However, the system will be designed to handle brief moments of poor connectivity gracefully. Core features like viewing the last loaded patient record will remain accessible even if the connection drops temporarily, with the data syncing once the connection is restored.

User Environment

The users of this system come from different backgrounds. Clinic staff range from receptionists with basic computer skills to experienced dermatologists who are familiar with clinical software. The system is designed to be welcoming and easy for all of these users, with clear labels, simple navigation, and helpful error messages.

Regulatory Environment

Uganda has data protection laws that govern how personal information, including health records, must be handled. The system is built in compliance with the Uganda Data Protection and Privacy Act. Patient consent is collected before any data is recorded, and patients have the right to request access to or deletion of their records at any time.

2.7 Conclusion

This chapter has laid out a clear picture of where Kampala Skin Clinic currently stands in terms of managing patient information, and why a new system is needed. The existing paper-based approach has served its purpose, but it is no longer sufficient for a growing clinic that sees a large number of patients every week. Lost records, missed appointments, unsafe data sharing, and the inability to track patient progress over time are problems that directly affect the quality of care.

The feasibility study confirmed that building this system is practical, affordable, and operationally sound. The database design ensures that patient data is stored in an organized, secure, and easy-to-access way. The requirements have been clearly defined, covering what the system must do and what it needs to run properly.

With this foundation in place, the next chapter will move on to the actual design of the system, including how the interface will look, how the different parts of the system will connect to each other, and how the user experience will be structured for both clinic staff and patients.

CHAPTER 3
SYSTEM DESIGN

3.0 Introduction

Software design is a step-by-step process where system requirements are turned into a clear plan for building the software. It helps in creating different representations of the system that guide development. These include the system architecture, data structures, user interfaces, and system components needed to make the system work.

Design is important because it helps improve the quality of the system before development begins. At this stage, errors can be identified and corrected early before coding starts, testing is done, and users begin interacting with the system.

This chapter focuses on the design of the **PearlCare Management System**. It begins with the system architecture, then explains the different modules of the system using simple diagrams and descriptions.

Different tools such as flowcharts and data flow diagrams are used to clearly show how the system works. These diagrams help both technical and non-technical users understand the system design.

The chapter ends with the database design, which includes the entity relationship diagram and table structures used to store and manage data in the system.

3.1 System Architecture

The PearlCare Management System follows a client-server architecture. In this model, the client (a web portal) sends requests to a central server, which processes those requests and sends back the appropriate response. This architecture was chosen because it keeps the data in one secure, central location while still allowing multiple users to access it from different devices at the same time.

The system is made up of three main layers. The presentation layer is what the user sees and interacts with, the business logic layer processes requests and applies rules, and the data layer stores and retrieves data from the database.

A web portal was developed so that the clinic staff can access the application from any computer or tablet with an internet browser, without needing to install anything.

The web portal communicates with a central backend server through a REST API. This API acts as the gateway between the user interfaces and the database, making sure that data is handled correctly and securely.

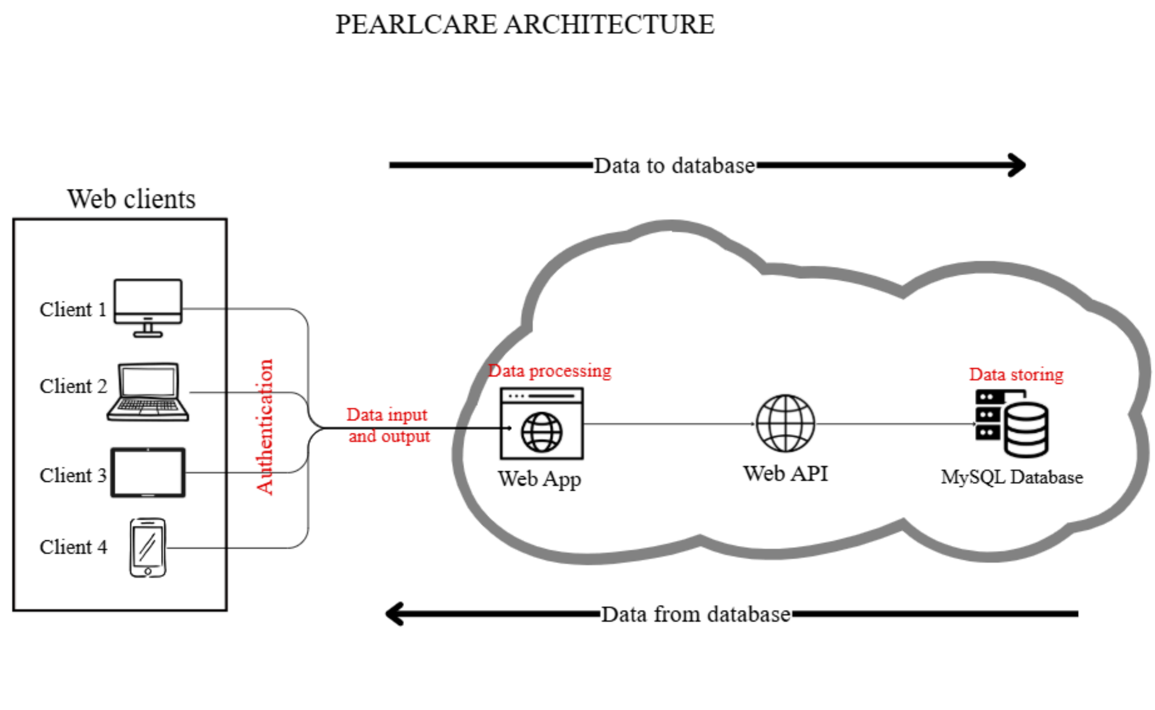


Figure 2:System Architecture.

3.2 Module Descriptions

The PearlCare Management System is divided into four core modules. Each module handles one specific area of the clinic's needs, and together they form a complete solution for skin clinic data management and patient care.

3.2.1 Patient Skin Health Data Collection

The first module is the data collection module. This is where all patient health information enters the system. Through the mobile application, a nurse or doctor can record the patient's current symptoms, note significant skin events such as a new rash or flare-up, update the patient's health profile, and upload photos of the skin conditions to track how they change over time.

3.2.2 Skin Health Data Sharing

The second module is the data sharing module. This module manages how patient data moves between different parties in the system, always in a controlled and secure way.

The doctors and nurses have access to the same modules which makes references and coordination more efficient at the clinic and offering of better services to the patients more possible.

The system also has a possibility of printing a report regarding the patients data which makes it possible for referrals to different health care facilities on request and proceeding treatments manageable.

3.2.3 Treatment Timeline Management

The third module manages active treatment plans. Dermatological treatments often run over weeks or months, involving a series of consultations, topical applications, or oral medications that must be taken on a strict schedule. This module helps both patients and clinic staff stay on top of these schedules.

Dermatologists can create treatment plans within the system and assign them to specific patients.

The module also keeps a timeline of all past treatments and consultations for each patient. This timeline is visible to both the nurses and the dermatologist, making it easy to see how a condition has responded to different treatments over time.

3.2.4 Diagnosis Support

The fourth module is the diagnosis support module, designed specifically for dermatologists and clinical staff. When a dermatologist has been granted access to a patient's records, this module presents that data in a clear and useful way to support clinical decision making.

The module displays visual trends showing how a patient's skin condition has changed over time. Photos uploaded are shown alongside symptom entries, giving the dermatologist a visual history that would be impossible to achieve with paper records alone.

The module also flags important patterns, such as a sudden worsening of symptoms after a new medication was introduced, or a recurring flare-up that happens at the same time every year. These markers help dermatologists spot things that might not be obvious when looking at individual records one by one.

3.3 Flow Charts

Flow charts have been used in engineering and software design for a long time. They are a straightforward way to show, step by step, how a particular scenario or process is handled within a system. In software design, flow charts make it easy to communicate how user actions lead to specific outcomes, making them a useful tool for both developers and non-technical stakeholders.

To understand how different scenarios are handled in the Kampala Skin Clinic Management System, a few key processes are illustrated below using flow charts. These include the patient registration process, the data sharing and access grant process.

3.3.1 Flow Chart

The system flow chart visually represents the logical sequence of operations a user undergoes when interacting with the platform. For any user except the administrator (Admin), this begins by the administrator adding them manually and the could then also assign a different user the rights and permissions to add other users.

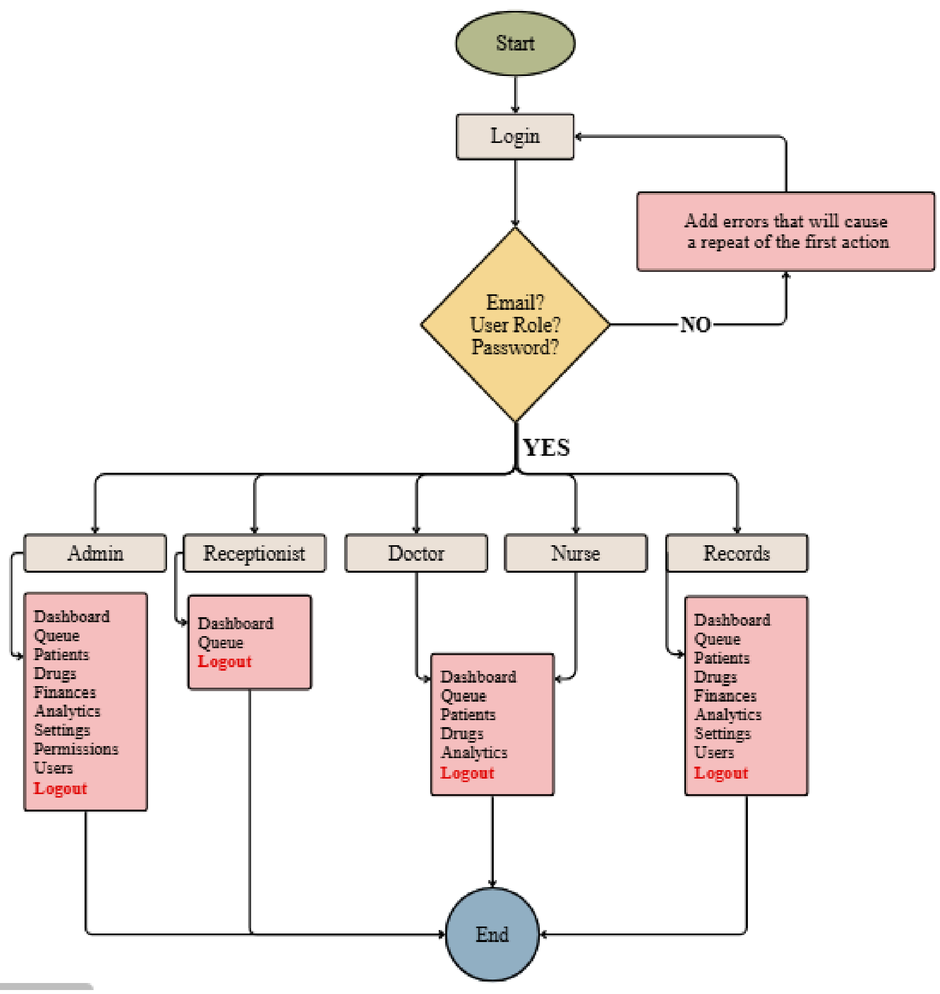


Figure 3: Flow chart of how different information flows in the system.

Adding a new user

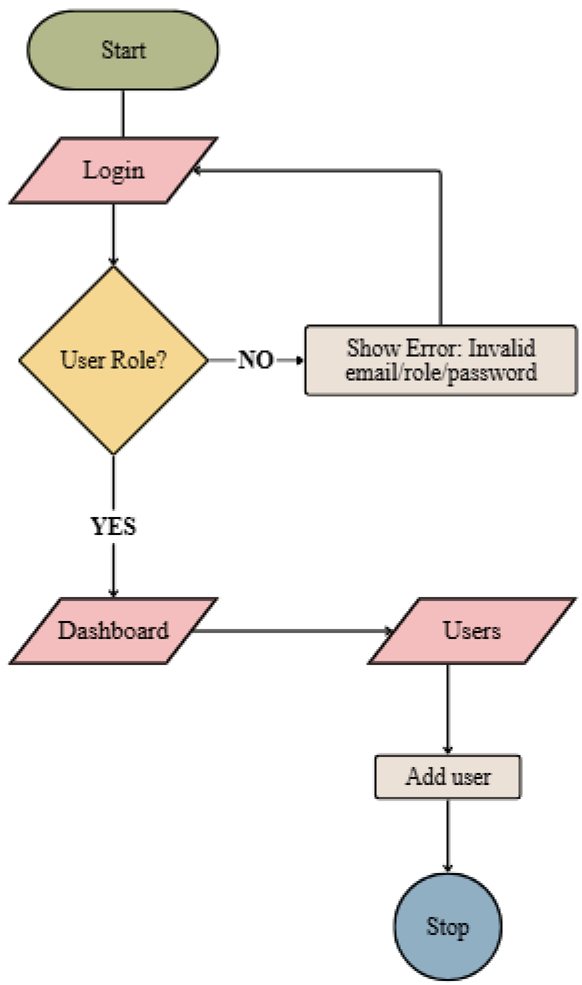


Figure 4: Flow chart of how a user is added into the system.

Adding permissions to user roles.

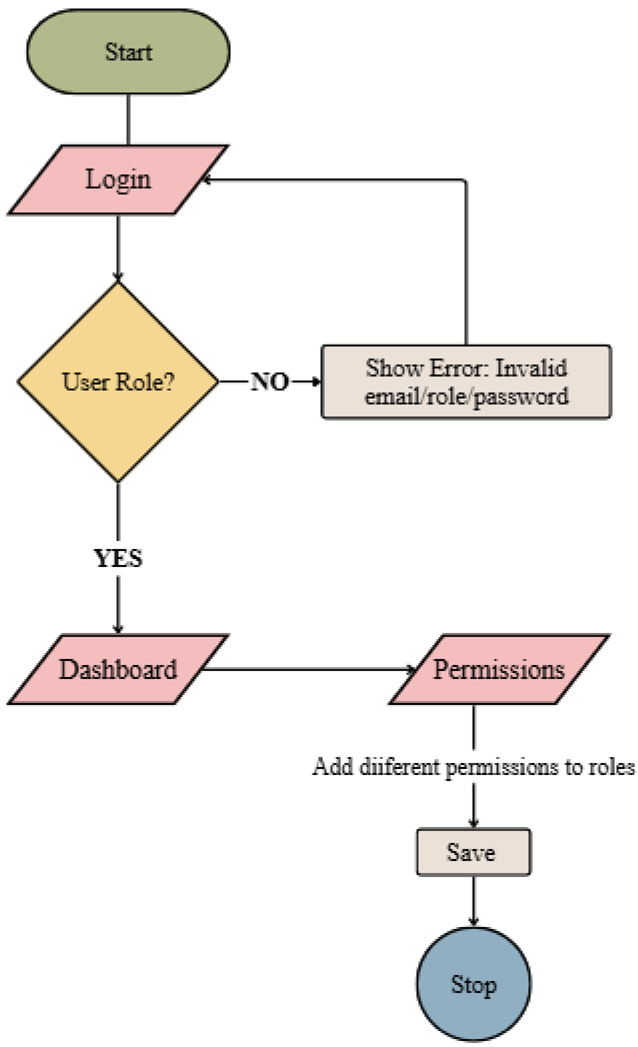


Figure 5: Flow chart of how a user is added into the system.

Printing a patient's report.

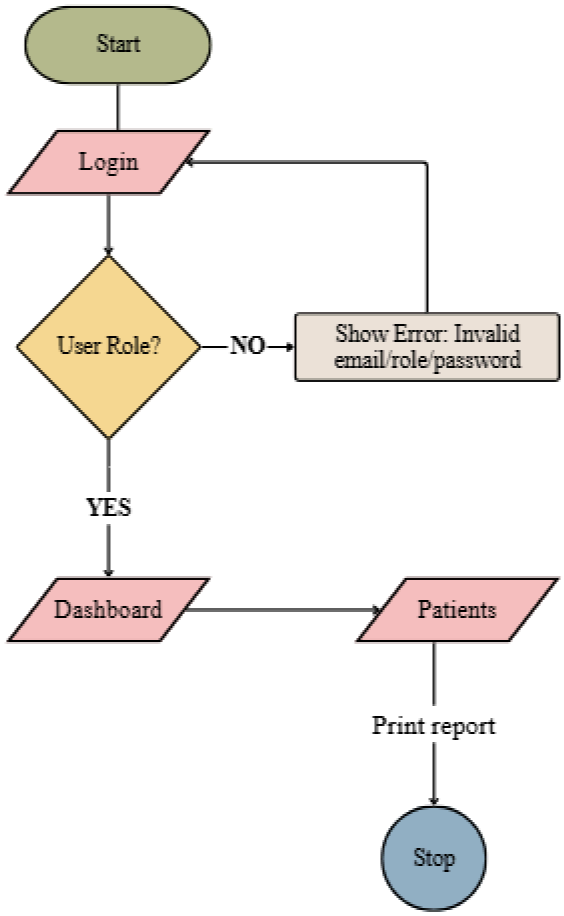


Figure 6: Flow chart of how a patient's form is printed.

3.4 Data Flow Diagrams

Data Flow Diagrams (DFDs) are used to show how data moves through a system. They are particularly useful for understanding how different parts of a system interact and what transformations data goes through as it passes between them.

The PearlCare Management System operates within a network of users and external services. At the highest level, the system receives input from clinic staff (through the web portal), processes that data according to the business rules, and stores or retrieves records from the central database.

Level 0 DFD (Context Diagram)

The context diagram shows the system as a single process at the centre, surrounded by the external entities that interact with it. These include dermatologists, receptionists, and the clinic's existing health reporting systems. Arrows show the direction in which data flows between the system and each entity.

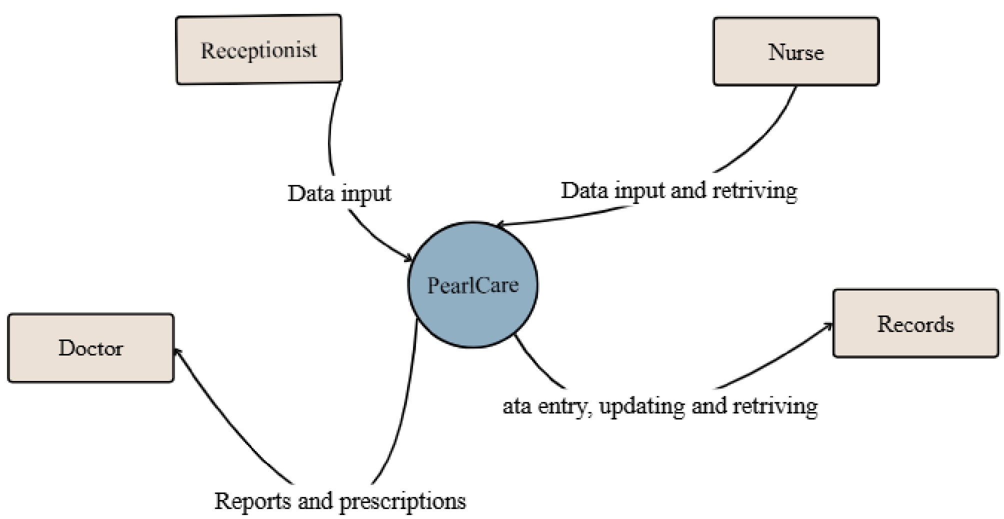


Figure 7: Data flow diagram of how information flows in level 0 in the system.

Level 1 DFD

The Level 1 diagram breaks the system down into its two main modules and shows how data flows between them. For example, data collected in the Data Collection module is stored in the database and then retrieved by the Diagnosis Support module when a dermatologist requests it.

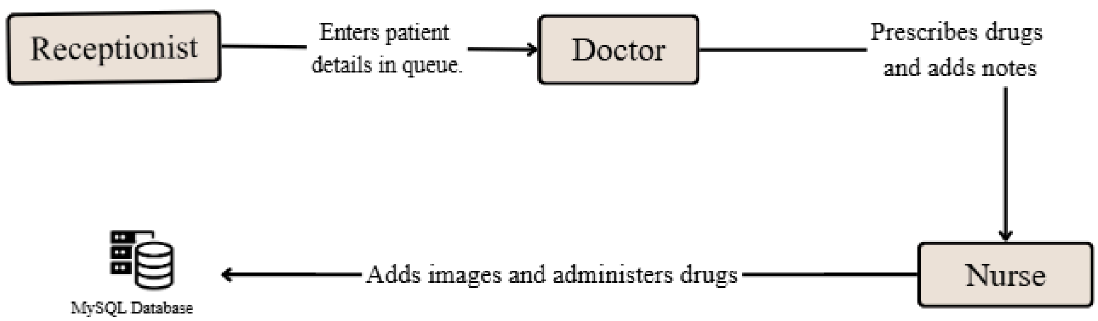


Figure 8: Data flow diagram of how information flows on level 1 in the system.

Level 2 DFD

The Level 2 diagrams zoom in on the individual modules from Level 1, breaking them down into their specific internal sub-processes. For example, within the Data Collection module, it shows the step-by-step flow of how patient vitals and images are captured, validated, and safely routed to the central database. Similarly, it details the exact sequence within the Diagnosis Support module, illustrating how that stored data is retrieved, analyzed, and turned into a final recommendation for the dermatologist.

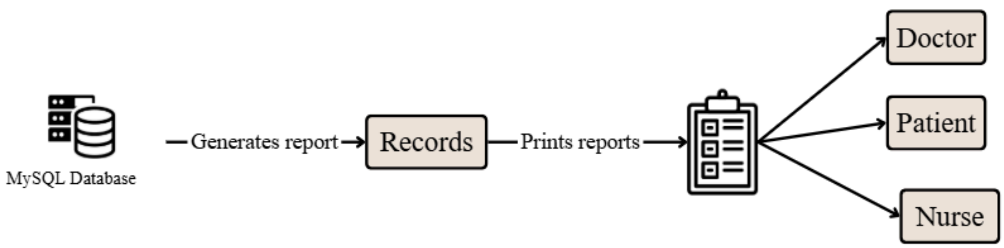


Figure 9: Data flow diagram of how information flows on level 2 in the system.

3.5. Database Design

The system uses a relational database built on MySQL. The database is designed to store all patient and clinical data in a structured way, making it fast to query and easy to maintain.

The Entity-Relationship (ER) diagram for the database shows the main entities and how they are connected to each other. The core entities are Patients, Users, Drugs, Prescriptions, Skin Condition Records, and Permissions. Each entity has a set of attributes and is linked to others through foreign keys.

Key constraints applied to the database include: every photo must be linked to a registered patient; a prescription cannot be saved without a linked consultation; an access grant(permission) must have both a nurse and a dermatologist linked to it; and all timestamps are recorded in UTC to avoid confusion across different time zones.

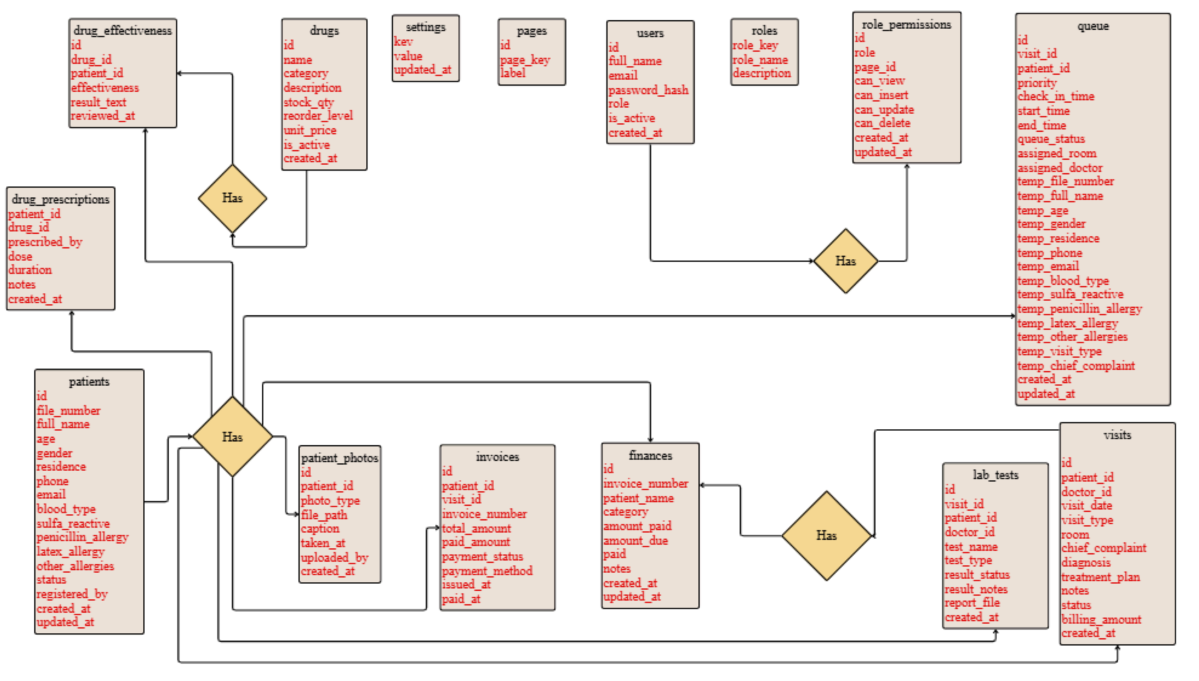


Figure 10: Database design of the system.

3.5.1 ER Diagram

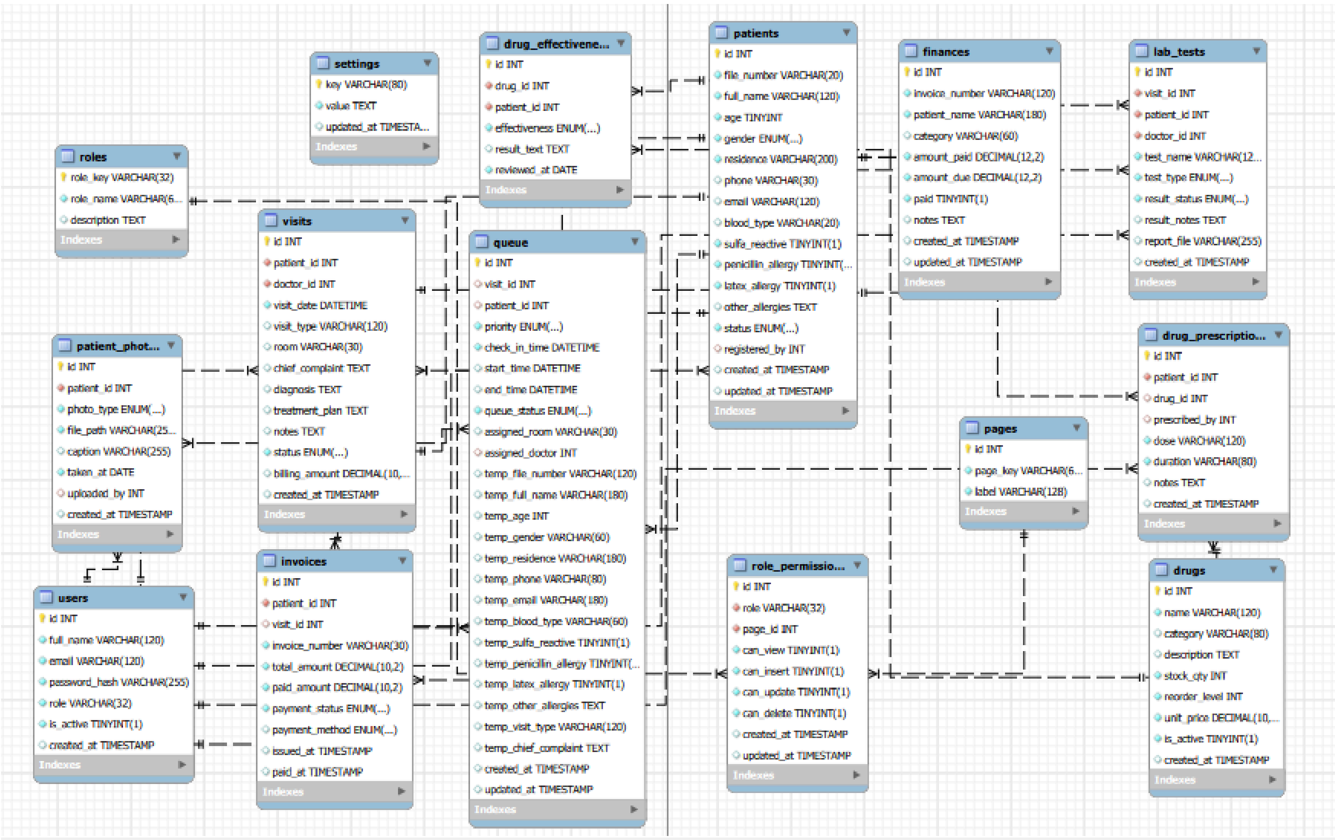


Figure 11: ER diagram of the system.

3.5.2 Table Design With Constraints

Below is the structural design of the primary database tables, highlighting their specific constraints and security measures.

TABLE_NAME	COLUMN_NAME	CONSTRAINT_NAME	REFERENCED_TABLE_NAME	REFERENCED_COLUMN_NAME	CONSTRAINT_
drug_effectiveness	drug_id	drug_effectiveness_ibfk_1	drugs	id	FOREIGN KEY
drug_effectiveness	patient_id	drug_effectiveness_ibfk_2	patients	id	FOREIGN KEY
drug_prescriptions	patient_id	drug_prescriptions_ibfk_1	patients	id	FOREIGN KEY
drug_prescriptions	drug_id	drug_prescriptions_ibfk_2	drugs	id	FOREIGN KEY
drug_prescriptions	prescribed_by	drug_prescriptions_ibfk_3	users	id	FOREIGN KEY
invoices	patient_id	invoices_ibfk_1	patients	id	FOREIGN KEY
invoices	visit_id	invoices_ibfk_2	visits	id	FOREIGN KEY
lab_tests	visit_id	lab_tests_ibfk_1	visits	id	FOREIGN KEY
lab_tests	patient_id	lab_tests_ibfk_2	patients	id	FOREIGN KEY
lab_tests	doctor_id	lab_tests_ibfk_3	users	id	FOREIGN KEY
patient_photos	patient_id	patient_photos_ibfk_1	patients	id	FOREIGN KEY
patient_photos	uploaded_by	patient_photos_ibfk_2	users	id	FOREIGN KEY
patients	registered_by	patients_ibfk_1	users	id	FOREIGN KEY
queue	visit_id	queue_ibfk_1	visits	id	FOREIGN KEY
queue	patient_id	queue_ibfk_2	patients	id	FOREIGN KEY
queue	assigned_doctor	queue_ibfk_3	users	id	FOREIGN KEY
role_permissions	role	role_permissions_ibfk_1	roles	role_key	FOREIGN KEY
role_permissions	page_id	role_permissions_ibfk_2	pages	id	FOREIGN KEY
visits	patient_id	visits_ibfk_1	patients	id	FOREIGN KEY
visits	doctor_id	visits_ibfk_2	users	id	FOREIGN KEY
drug_effectiveness	id	PRIMARY	NULL	NULL	PRIMARY KEY
drug_prescriptions	id	PRIMARY	NULL	NULL	PRIMARY KEY
drugs	id	PRIMARY	NULL	NULL	PRIMARY KEY
finances	invoice_number	invoice_number	NULL	NULL	PRIMARY KEY
finances	id	PRIMARY	NULL	NULL	PRIMARY KEY
invoices	invoice_number	invoice_number	NULL	NULL	PRIMARY KEY
invoices	id	PRIMARY	NULL	NULL	PRIMARY KEY
lab_tests	id	PRIMARY	NULL	NULL	PRIMARY KEY
pages	page_key	page_key	NULL	NULL	PRIMARY KEY
pages	id	PRIMARY	NULL	NULL	PRIMARY KEY
patient_photos	id	PRIMARY	NULL	NULL	PRIMARY KEY
patients	file_number	file_number	NULL	NULL	PRIMARY KEY
patients	id	PRIMARY	NULL	NULL	PRIMARY KEY
queue	id	PRIMARY	NULL	NULL	PRIMARY KEY
role_permissions	id	PRIMARY	NULL	NULL	PRIMARY KEY
role_permissions	role	unique_role_page	NULL	NULL	PRIMARY KEY
role_permissions	page_id	unique_role_page	NULL	NULL	PRIMARY KEY
roles	role_key	PRIMARY	NULL	NULL	PRIMARY KEY
settings	key	PRIMARY	NULL	NULL	PRIMARY KEY
users	email	email	NULL	NULL	PRIMARY KEY
users	id	PRIMARY	NULL	NULL	PRIMARY KEY
visits	id	PRIMARY	NULL	NULL	PRIMARY KEY

3.6 Conclusion

System design is a vital step in the journey from an idea to a working product. This chapter walked through the major design decisions for the PearlCare Management System, starting with the overall client-server architecture, then diving into each of the four modules, and illustrating how key scenarios work using flow charts and DFDs.

The design ensures that the system is secure, usable, and scalable. The modular structure means that each part of the system can be improved or extended in the future without affecting the others. The next chapter will show how this design was turned into actual working software.

CHAPTER 4

SYSTEM DEVELOPMENT

4.0 Introduction

System development, commonly known as system implementation, is the process of turning a design into a working software product. This can be done in different ways: configuring an off-the-shelf system, assembling existing software components, or writing code from scratch. For the Kampala Skin Clinic Management System, the system was built primarily through coding, because no existing off-the-shelf product fully addressed the specific needs of a dermatology clinic in the Ugandan context.

Different parts of the system were built using a combination of programming languages and frameworks. The web portal for clinic staff was built using PHP and the database runs on MySQL.

This chapter covers all the key activities involved in building the system. It starts with the implementation plan and schedule, then presents the user interface through screenshots of key screens, shows selected code snippets to demonstrate the technical approach taken, and concludes with a discussion of the testing that was done to validate the system.

4.1 Implementation Plan

Planning is a critical part of any software development project. A well-thought-out plan keeps the team focused, helps identify risks early, and provides a clear measure of progress throughout the project.

The implementation of the PearlCare Management System was planned to run over approximately two months, from 1 May 2025 to 30 June 2025. An additional two-week buffer was included to account for any unexpected delays. The table below shows the detailed project schedule:

Activity	Start Date	End Date	Deliverables
Orientation	30 March 2026	31 March 2026	Guidance on how to do different case studies and develop the project.
Software Requirements Specification	1 May 2026	10 May 2026	Requirements document, tasks and actions list
Requirements Analysis	11 April 2026	17 April 2026	Validated requirements specification
System Design	18 April 2026	25 April 2026	Architectural design, database design, diagrams and models
Implementation (Coding)	26 April 2026	9 May 2026	UI design, backend and frontend code, operational software ready for testing
Testing and Deployment	10 May 2026	23 May 2026	Manual testing report, user acceptance testing report
Miscellaneous / Buffer	24 May 2026	30 May 2026	Final adjustments and handover

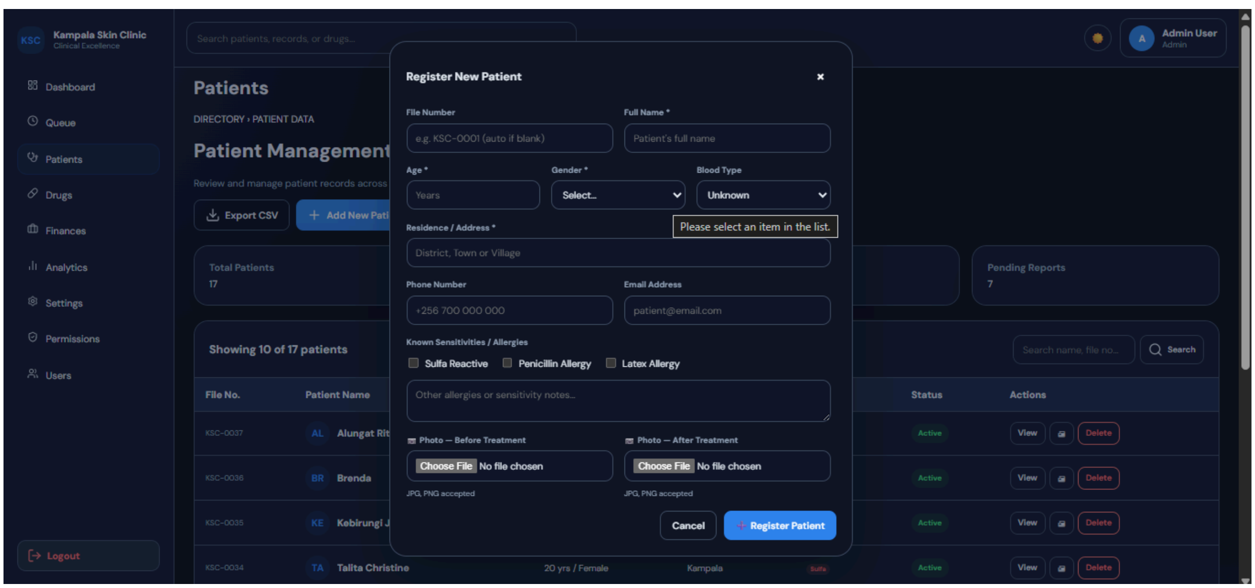
4.2 Screenshots

The PearlCare Management System was designed to be visually clean, professional, and easy to use. The interface uses a consistent colour scheme, clear labels, and simple navigation so that users can complete their tasks quickly without confusion.

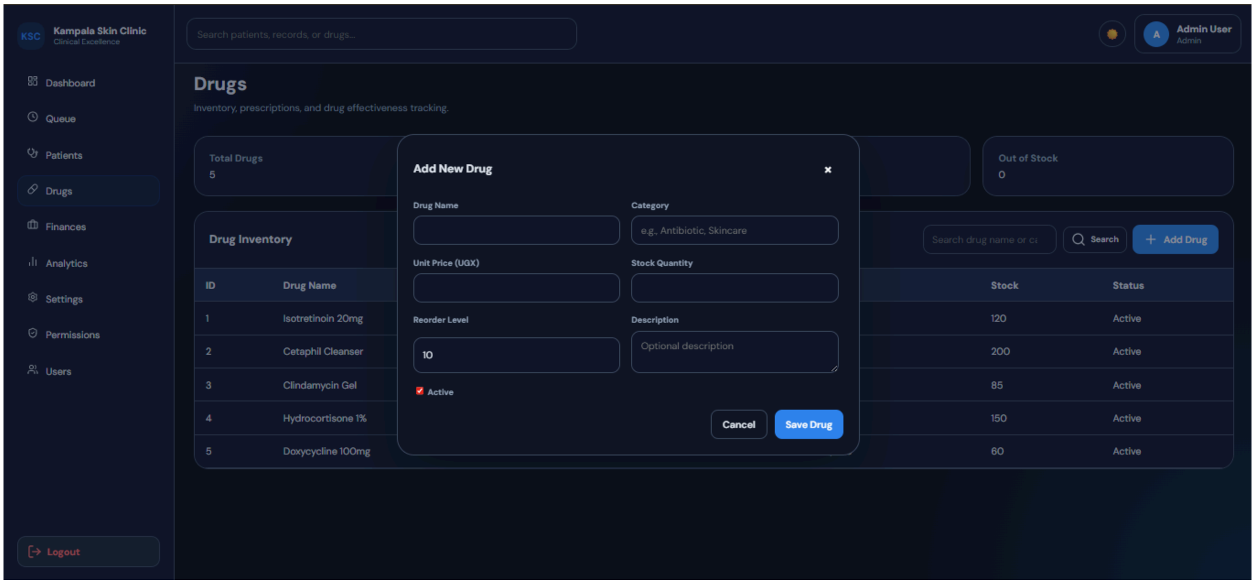
4.2.1 Input Design

Input screens were designed to be straightforward and guide the user through what is needed. Every form has clear labels, helpful placeholder text, and validation messages that appear immediately if something is entered incorrectly. The colour teal was chosen as the primary colour for the system, reflecting a clean, medical feel. The font used throughout is Sans-serif, which is modern and easy to read on screens of all sizes.

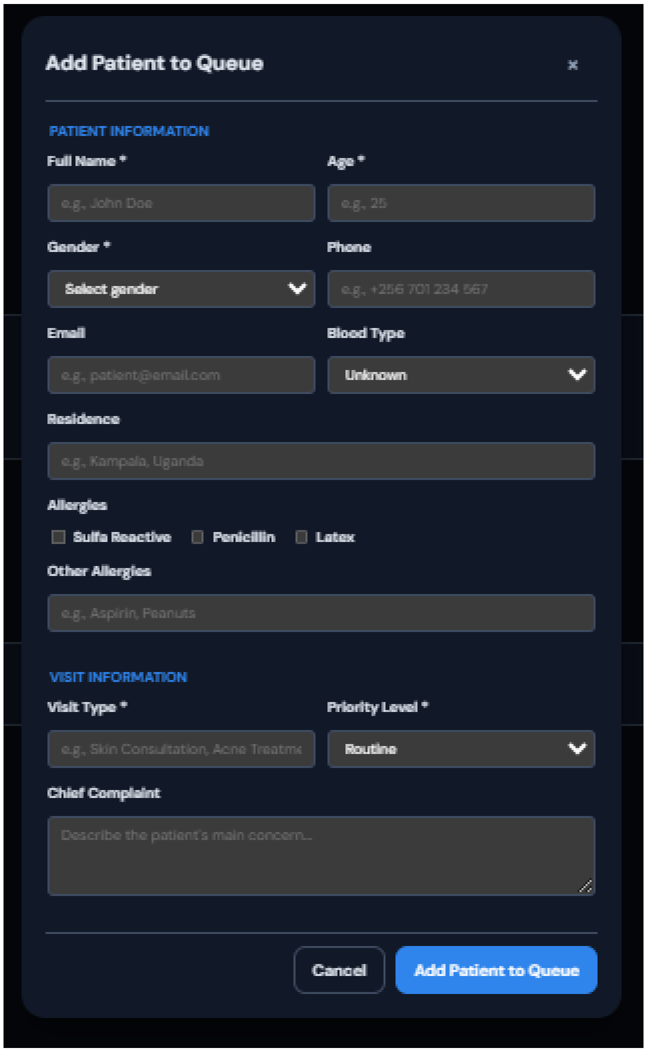
Key input screens include the patient registration form, the users registration form, and the consultation notes form. Each screen was tested with real users to make sure the layout made sense and the actions were intuitive.



New patient registration form.



New patient registration form.

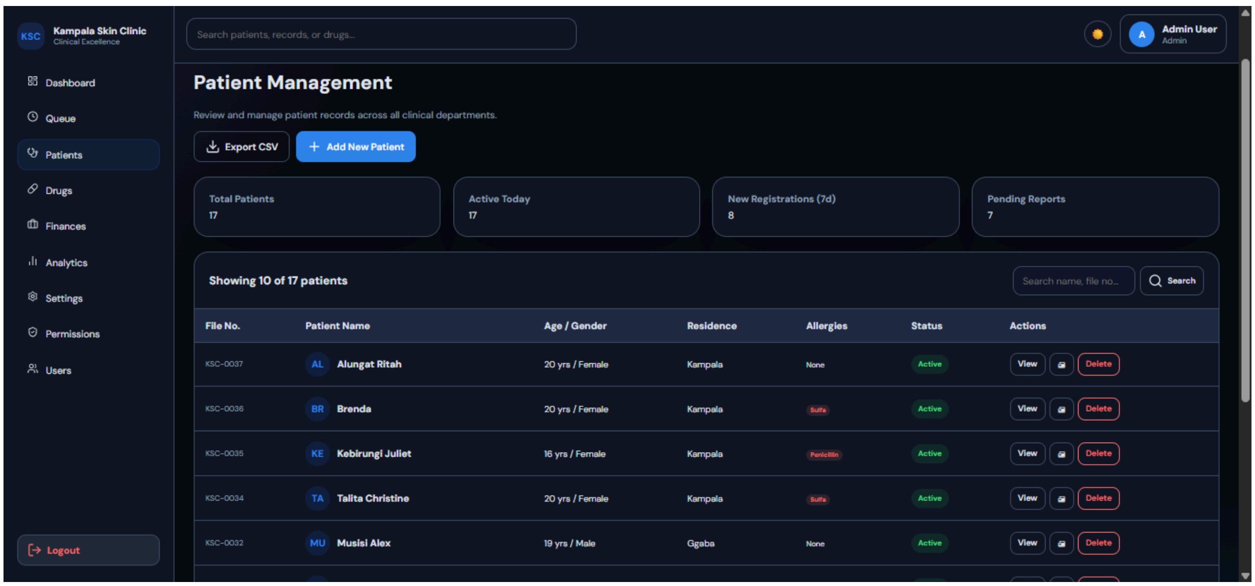


New patient registration form into queue.

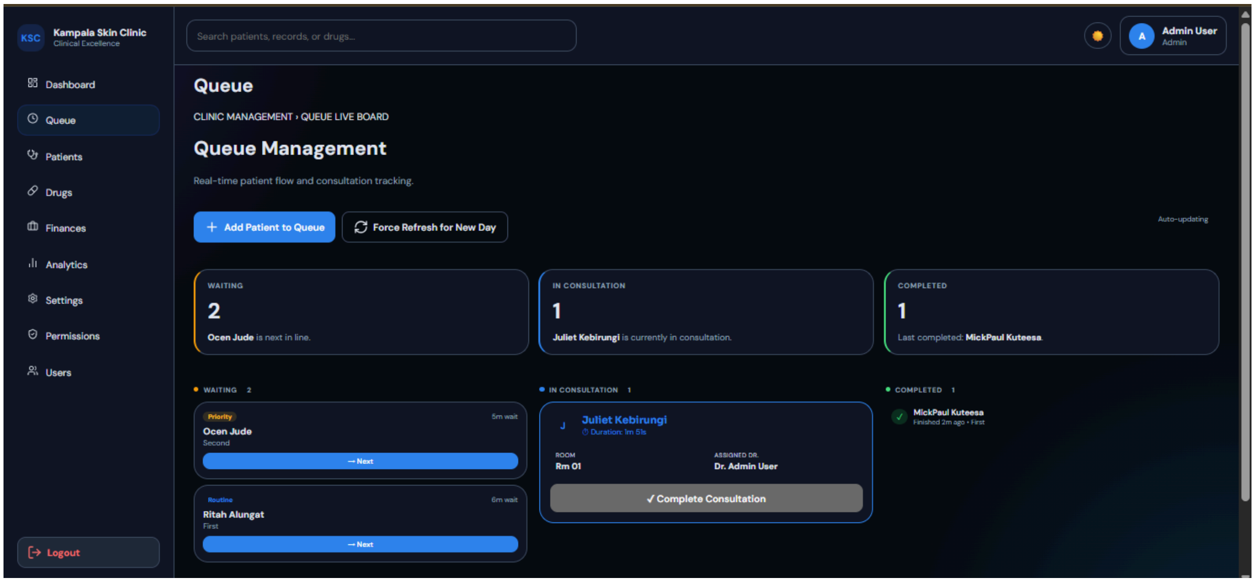
4.2.2 Output Design

Output screens present information in a clear and organized way. The dermatologist's dashboard shows a summary of each patient's most recent condition and any flags raised by the diagnosis support module.

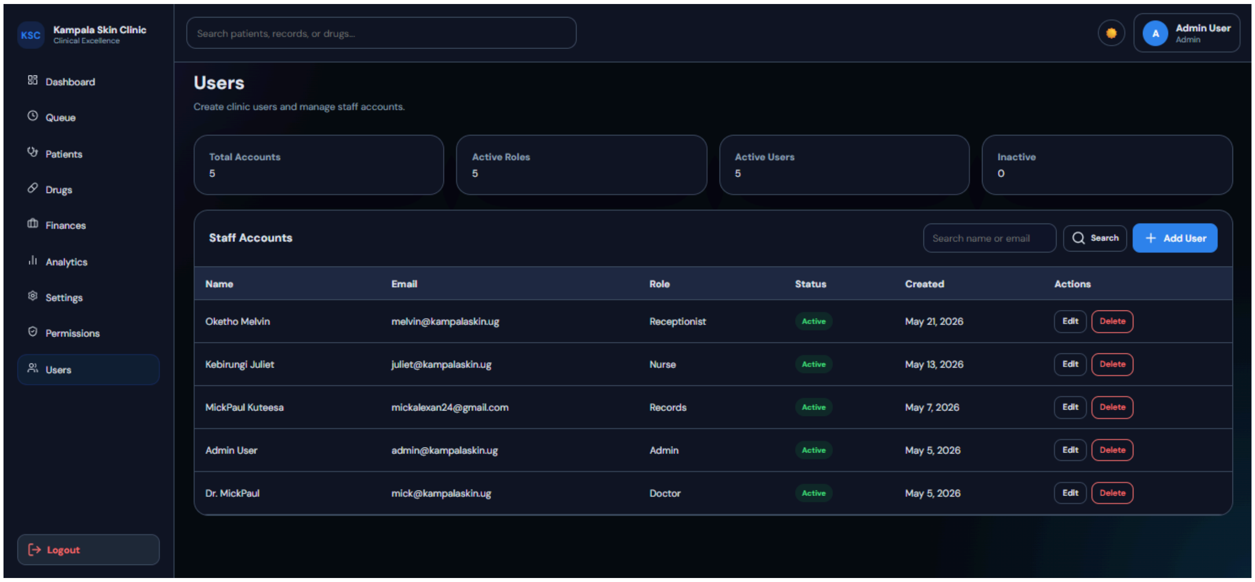
All output screens follow the same visual style as the input screens for consistency. CSS3 was used to style the web portal. Charts and graphs used in the diagnosis support module are rendered using a lightweight charting library, making skin condition trends easy to read and interpret at a glance.



Displaying patients in the system database.



Display patients in the queue.



Displaying users in the system.

4.3 Coding

The PearlCare Management System was built using a combination of technologies that were selected based on their reliability, security, ease of use, and strong community support.

Web Portal (PHP): The clinic staff portal was built with PHP that supports both server-side rendering and static generation. This ensures fast load times and a smooth experience for receptionists and dermatologists working from clinic computers.

Database (MySQL): MySQL was used as the primary database. It stores all patient records, consultations, and prescriptions. All sensitive fields, such as passwords, are hashed before storage. The database is hosted on a secure cloud server with automated daily backups.

The development environment made use of Visual Studio Code as the primary code editor, Git and GitHub for version control and collaboration, and Postman for testing API endpoints during development.

Code Sample 1: Role Permissions Matrix Definition

The following SQL declaration defines the schema for the `role_permissions` table. This table acts as the structural foundation for the application's Role-Based Access Control (RBAC) system, as it maps system roles to individual application pages and dictates the granular, operation-level permissions granted to users.

```
CREATE TABLE IF NOT EXISTS role_permissions (
    id INT AUTO_INCREMENT PRIMARY KEY,
    role VARCHAR(32) NOT NULL,
    page_id INT NOT NULL,
    can_view TINYINT(1) NOT NULL DEFAULT 0,
    can_insert TINYINT(1) NOT NULL DEFAULT 0,
    can_update TINYINT(1) NOT NULL DEFAULT 0,
    can_delete TINYINT(1) NOT NULL DEFAULT 0,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
    FOREIGN KEY (role) REFERENCES roles(role_key) ON DELETE CASCADE,
    FOREIGN KEY (page_id) REFERENCES pages(id) ON DELETE CASCADE,
    UNIQUE KEY unique_role_page (role, page_id)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

The **`UNIQUE KEY`** constraint on **`(role, page_id)`** is enforced directly by the database engine to guarantee that no duplicate or conflicting permission records can exist for a specific role on a specific page. Furthermore, the **`FOREIGN KEY`** declarations utilize **`ON DELETE CASCADE`** to maintain strict referential integrity. The result is that if a role is deleted or a page is removed from the system, the database engine itself automatically cleans up all associated permission configurations. This provides a rigid, fail-safe authorization matrix that prevents orphaned access rules and allows the application's backend middleware to reliably authorize user actions before executing any CRUD operations.

Code Sample 2: Authentication and Session Management

The following PHP script defines the backend logic for the application's user authentication gateway (**`login.php`**). This component is critical to system security, as it validates incoming user credentials, securely verifies cryptographic password hashes, and initializes the user session before granting access to the protected dashboard.

```
<?php
require_once 'config.php';
```

```
// If already logged in, skip the login page
if (!empty($_SESSION['user_id'])) {
    header('Location: dashboard.php');
    exit;
}

$error = ''; // Will hold any error message to show the user

// ---- Handle form submission -------------------------------
if ($_SERVER['REQUEST_METHOD'] === 'POST') {

    // Get what the user typed (trim removes extra spaces)
    $email    = trim($_POST['email']    ?? '');
    $password =      $_POST['password'] ?? '';
    $role     = trim($_POST['role']     ?? '');
    error_log(date('Y-m-d H:i:s') . " - POST received: email=" .
$email . ", role=" . $role . "\n", 3, LOG_FILE);

    if ($email === '' || $password === '' || $role === '') {
        $error = 'Please enter your email, password, and role.';
    } else {
        // Look up the user by email and role
        $conn = db();
        $user = null;
        $stmt = $conn->prepare("SELECT * FROM users WHERE email = ?
AND role = ? AND is_active = 1 LIMIT 1");
        if ($stmt === false) {
            error_log(date('Y-m-d H:i:s') . " - DB prepare failed: "
. $conn->error . "\n", 3, LOG_FILE);
            $error = 'Unable to process login right now. Please try
again later.';
        } else {
            $stmt->bind_param("ss", $email, $role);
            $stmt->execute();
            $result = $stmt->get_result();
            if ($result === false) {
                error_log(date('Y-m-d H:i:s') . " - DB get_result
failed: " . $stmt->error . "\n", 3, LOG_FILE);
                $error = 'Unable to process login right now. Please
```

```
try again later.';
            } else {
                $user = $result->fetch_assoc();
            }
            $stmt->close();
        }

        if ($user) {
            // User found, now verify password
            if (password_verify($password, $user['password_hash'])) {
                $_SESSION['user_id'] = $user['id'];
                $_SESSION['user']    = $user;
                // Debug: Log successful login and session data
                error_log(date('Y-m-d H:i:s') . " - Login successful
for user: " . $email . "\n", 3, LOG_FILE);
                error_log(date('Y-m-d H:i:s') . " - Session user_id
set to: " . $_SESSION['user_id'] . "\n", 3, LOG_FILE);
                error_log(date('Y-m-d H:i:s') . " - Session data: " .
json_encode($_SESSION['user']) . "\n", 3, LOG_FILE);

                // Make sure session is written to disk before
redirecting
                session_write_close();

                // Verify headers haven't been sent
                if (headers_sent($file, $line)) {
                    error_log(date('Y-m-d H:i:s') . " - Headers
already sent in $file:$line - redirect failed!", 3, LOG_FILE);
                    $error = 'Error: Headers already sent. Please
contact administrator.';
                } else {
                    header('Location: dashboard.php');
                    exit;
                }
            } else {
                // Password doesn't match
                error_log(date('Y-m-d H:i:s') . " - Password
verification failed for user: " . $email . "\n", 3, LOG_FILE);
                error_log(date('Y-m-d H:i:s') . " - Expected hash: "
```

```
. $user['password_hash'] . "\n", 3, LOG_FILE);
                $error = 'Incorrect password. Please try again.';
            }
        } else {
            // User not found
            error_log(date('Y-m-d H:i:s') . " - User not found: 
email=" . $email . ", role=" . $role . "\n", 3, LOG_FILE);
            $error = 'Incorrect email, role, or password. Please try 
again.';
        }
    }
}
```

The authentication flow utilizes parameterized SQL queries (`$conn->prepare`) to strictly prevent SQL injection attacks when querying the database by email and role. When a matching, active user is found, the script employs PHP's native `password_verify()` function to securely compare the submitted plaintext password against the stored cryptographic hash. Detailed error logging is implemented throughout the process to aid administrators in auditing failed attempts and diagnosing server issues without exposing sensitive system details to the end-user. Upon successful verification, the script writes the user's data to the session variable, explicitly saves the session data to disk using `session_write_close()` to prevent race conditions, and safely redirects the user to the application dashboard.

Code Sample 3: Live Patient Queue Management Logic

The following PHP script serves as the data controller for the clinic's live queue management dashboard (`queue.php`). This module is a critical operational component, acting as the real-time traffic controller for the facility by querying the database to segment patient traffic into three distinct, actionable states: Waiting, In Consultation, and Completed.

```
<?php
require_once 'config.php';
requireLogin();

$conn = db();

// Handle daily queue reset if needed
$lastClearDate = $_SESSION['queue_last_clear_date'] ?? null;
$today = date('Y-m-d');
```

```
if ($lastClearDate !== $today) {
    // First time loading queue today, initialize
    $_SESSION['queue_last_clear_date'] = $today;
}

// ---- Get Waiting patients (ordered by priority, then check-in
time) ----
$waitingResult = $conn->query("
    SELECT q.id, q.patient_id, q.visit_id, q.priority,
q.check_in_time,
           COALESCE(p.full_name, q.temp_full_name) AS full_name,
           COALESCE(p.age, q.temp_age) AS age,
           COALESCE(v.visit_type, q.temp_visit_type) AS visit_type,
           COALESCE(v.chief_complaint, q.temp_chief_complaint) AS
chief_complaint
    FROM queue q
    LEFT JOIN patients p ON p.id = q.patient_id
    LEFT JOIN visits v ON v.id = q.visit_id
    WHERE q.queue_status = 'Waiting' AND DATE(q.check_in_time) =
CURDATE()
    ORDER BY FIELD(q.priority, 'Urgent', 'Priority', 'Routine'),
q.check_in_time ASC
");

// ---- Get Patients currently in consultation ----
$consultResult = $conn->query("
    SELECT q.id, q.patient_id, q.visit_id, q.start_time,
q.assigned_room, q.assigned_doctor,
           COALESCE(p.full_name, q.temp_full_name) AS full_name,
           COALESCE(v.visit_type, q.temp_visit_type) AS visit_type,
           u.full_name AS doctor_name
    FROM queue q
    LEFT JOIN patients p ON p.id = q.patient_id
    LEFT JOIN visits v ON v.id = q.visit_id
    LEFT JOIN users u ON u.id = q.assigned_doctor
    WHERE q.queue_status = 'In Consultation' AND
DATE(q.check_in_time) = CURDATE()
    ORDER BY q.start_time ASC
```

```
");

// ---- Get Completed today ----
$completedResult = $conn->query("
    SELECT q.id, q.patient_id, q.visit_id, q.end_time,
           COALESCE(p.full_name, q.temp_full_name) AS full_name,
           COALESCE(v.visit_type, q.temp_visit_type) AS visit_type
    FROM queue q
    LEFT JOIN patients p ON p.id = q.patient_id
    LEFT JOIN visits v ON v.id = q.visit_id
    WHERE q.queue_status = 'Completed' AND DATE(q.check_in_time) =
CURDATE()
    ORDER BY q.end_time DESC
    LIMIT 10
");

// Collect into arrays for easy use in the view
$waiting   = $waitingResult   ?
$waitingResult->fetch_all(MYSQLI_ASSOC)   : [];
$consult   = $consultResult   ?
$consultResult->fetch_all(MYSQLI_ASSOC)   : [];
$completed = $completedResult ?
$completedResult->fetch_all(MYSQLI_ASSOC) : [];

// Get current user info
$currentUser = $conn->query("SELECT * FROM users WHERE id = " .
$_SESSION['user_id'])->fetch_assoc();
$currentUserName = $currentUser['full_name'] ?? 'Doctor';

$pageTitle  = 'Queue';
$activePage = 'Queue';
require_once 'nav.php';
?>
```

The script efficiently structures the patient flow using three targeted SQL queries, restricting all results to the current date via `CURDATE()`. A standout architectural feature is the strategic use of the SQL `COALESCE` function across the `LEFT JOIN` statements. This allows the system to seamlessly handle a hybrid data model: it prefers data from fully registered patients (pulling from the normalized `patients` and `visits` tables) but gracefully falls back to temporary data stored directly on the `queue` record for unregistered walk-ins.

Furthermore, the medical triage logic is enforced directly at the database level rather than in PHP. By utilizing `ORDER BY FIELD(q.priority, 'Urgent', 'Priority', 'Routine')`, the database engine itself guarantees that severe cases automatically bypass routine visits in the returned dataset. This ensures a robust, fail-safe prioritization mechanism that drives the frontend view accurately every time the dashboard auto-refreshes.

4.4 Testing

Testing is an essential part of software development. It validates that the system does what it is supposed to do, identifies bugs before they reach real users, and builds confidence in the quality and reliability of the product.

Three main types of testing were carried out for the Kampala Skin Clinic Management System:

4.4.1 Unit Testing

Unit tests were written to verify the correctness of individual pieces of logic within the system. For example, tests were written to confirm that the appointment conflict checker correctly identified overlapping slots, that the symptom severity calculator produced the right output, and that access grant rules were applied correctly. Unit tests were run automatically every time new code was pushed to the repository, ensuring that changes did not accidentally break existing functionality.

4.4.2 Integration Testing

Integration tests were conducted to ensure that the distinct modules of the PearlCare Management System communicated and functioned together seamlessly. While unit testing verified individual components, integration testing validated the data flow across the system's overall architecture. For instance, scenarios were tested to verify that when a receptionist scheduled a new appointment, the database updated accurately and the dermatologist's schedule dashboard reflected the changes in real-time. Additionally, the interactions between the frontend patient records module and the backend database were rigorously evaluated to ensure that doctors could securely retrieve, update, and save diagnostic notes without data loss or corruption. This phase was crucial for identifying and resolving hidden communication defects between the system's APIs and user interfaces.

4.4.3 Security Testing

Given that the system handles sensitive patient health data, security testing was taken very seriously. Tests were carried out to check that password hashing was working correctly, that users could not access records they were not authorized to see, and that API endpoints rejected requests without valid authentication. Strong password rules were enforced, and SQL injection and cross-site scripting vulnerabilities were tested and addressed.

4.4.4 User Acceptance Testing

The system was handed to a selected group of users, including two dermatologists, two receptionists, who were asked to complete specific tasks and then share their feedback. The sessions were observed and the users were interviewed afterwards about what worked well, what was confusing, and what could be improved. All major issues identified during this phase were addressed before final deployment. The testing confirmed that the system met its requirements and that users were satisfied with the experience.

4.5. Conclusion

System development brought the design of the PearlCare Management System to life. Dermatologists, receptionists, and patients at Kampala Skin Clinic now have access to a software system for securely collecting, managing, and sharing skin health data. The system consists of a web portal for clinic staff.

This chapter provided a clear picture of how the system was built, from the implementation schedule and user interface design through to the coding approach and the testing that was done to validate the final product. The next chapter will reflect on how the system performs in the real world, what opportunities it opens up, and what challenges still need to be addressed.

CHAPTER 5

RECOMMENDATIONS AND CONCLUSION

5.0 Introduction

The PearlCare Management System was developed and delivered successfully at Kampala Skin Clinic in Kampala, Uganda. This final chapter reflects on the journey from the beginning of the project to its completion. It compares how things worked before the system was introduced with how they work now, highlights the opportunities the system has created, discusses the challenges that came up along the way, and explains the key development decisions that were made.

5.1 Discussion

The main outcome of this project is a working software system that is now being used by clinic staff and patients at Kampala Skin Clinic. To understand the value that the system has added, it is useful to compare the old way of doing things with the new one.

5.1.1 Comparison with the Previous System

Before this project, Kampala Skin Clinic relied heavily on paper patient cards and basic spreadsheets to manage patient records. When a patient visited the clinic, their details were written by hand into a physical file. These files were stored in a cabinet and retrieved manually at each visit. Appointment scheduling was done over the phone or in person, and reminders depended entirely on the patient remembering their next visit.

Treatment notes, prescriptions, and follow-up instructions were written by hand and given to the patient on paper. There was no reliable way to track a patient's skin condition over time, and photos of skin lesions were rarely taken because there was no system to store them against a patient record.

The new system has changed this in several meaningful ways. Patient records are now stored digitally in a secure central database, accessible to authorized staff in seconds. Visual history of patients' data conditions that the dermatologist can review before and during each consultation.

Appointment reminders are now sent automatically, reducing missed appointments. Prescriptions are issued digitally and visible to both the nurse and the dermatologist. Data sharing, which previously required handing over a physical file, now happens instantly through a permission-based access system that the patient fully controls.

5.1.2 Opportunities

The introduction of the Kampala Skin Clinic Management System has opened up several meaningful opportunities for better care and service delivery.

- **Better long-term condition management:** Patients with chronic skin conditions like eczema, psoriasis, or vitiligo can now be tracked consistently over months and years. Dermatologists get a much fuller picture of how a condition behaves over time, which leads to more informed treatment decisions.
- **More open patient communication:**
- **Improved medication adherence:** The treatment reminder feature helps patients stick to their medication schedules. This is particularly important for conditions like acne or fungal infections that require consistent daily treatment to be effective.
- **Emergency care support:** The emergency access feature allows clinical staff to quickly retrieve key health details for a patient who arrives unable to communicate. Known allergies, blood type, and current medications can be accessed instantly, potentially preventing dangerous errors in emergency treatment.
- **Data for public health planning:** As more patients use the system, the clinic builds a growing dataset of dermatological conditions in Kampala. This data can be anonymized and used by public health authorities to understand the prevalence of skin conditions in the city and plan health services accordingly.

5.1.3 Challenges

As with any new system, the Kampala Skin Clinic Management System has also brought its share of challenges.

- **Digital literacy:**
- **Data accuracy:** Because patients record their own symptoms without clinical supervision, there is a risk that some entries may be inaccurate or incomplete. This was partially addressed by designing the symptom input screens to guide

patients through a structured recording process, but it is something that clinic staff must stay mindful of.

- **Security of patient-held data:** Moving health data from the clinic's own systems onto patients' personal smartphones introduces new security risks. If a patient's phone is lost or stolen, there is a risk that their health data could be accessed by someone else. The app uses PIN and biometric lock features to reduce this risk, but users need to be reminded to keep these enabled.
- **Resistance from some staff:** A small number of clinic staff expressed reservations about trusting patient-recorded symptoms in the same way they would trust a record taken by a clinical officer. This is an understandable concern, and it was addressed by making it clear that patient-recorded data is a supplement to, not a replacement for, clinical assessment.

5.1.4 Development Reflections

To deliver the Kampala Skin Clinic Management System, a structured development schedule was followed as described in Chapter 4. Several technology choices were made to meet the specific requirements of the system.

The web portal was built with PHP because clinic staff work primarily from desktop computers at the clinic, and a web-based portal is easier to update and maintain than a desktop application. It also means that no installation is required on clinic computers.

5.2 Conclusion

This chapter brought together the key reflections from the entire project. It compared the clinic's old paper-based approach with the new digital system, highlighted the real opportunities the system creates for better skin health management in Kampala, and was honest about the challenges that remain.

The Kampala Skin Clinic Management System has demonstrated that a well-designed, patient-centred digital health system can make a meaningful difference in a busy specialist clinic. It has improved how records are kept, how appointments are managed, and how patients engage with their own health. With continued adoption and a few improvements over time, the system has the potential to become a model for other specialist clinics in Uganda.

5.3 Summary

This document has walked through the complete development journey of the Kampala Skin Clinic Management System from start to finish.

Chapter One introduced the project by exploring the background of Kampala Skin Clinic, the state of health information systems in Uganda, the specific problem this project set out to solve, and the objectives, scope, significance, and limitations of the work.

Chapter Two provided a system analysis, covering the existing paper-based system and its shortcomings, the feasibility study that confirmed the project was practical, and the detailed requirements for the new system.

Chapter Three tackled the system design, describing the client-server architecture, the four core modules, key user scenarios through flow charts and DFDs, and the relational database structure.

Chapter Four covered the implementation of the system, including the development schedule, the user interface design, the technology stack and code, and the testing that was done to validate the product.

Chapter Five closed the document with a discussion comparing the old and new systems, the opportunities the new system creates, the challenges it presents, and the development decisions that shaped it.

Overall, this project successfully delivered a working skin clinic management system that empowers clinic staff to deliver high-quality dermatological care. The system is now in active use at Kampala Skin Clinic and is already showing positive results in record accuracy, appointment attendance, and patient engagement.

FUTURE WORK

The PearlCare Management System has laid a strong foundation for digital dermatological health management in Kampala. However, there is meaningful room for the system to grow and improve in the future.

One of the most exciting opportunities is the incorporation of Artificial Intelligence (AI) into the diagnosis support module. AI models trained on large datasets of dermatological images have shown promising results in identifying common skin conditions like melanoma, psoriasis, and eczema from photos. Integrating such a model into the system could give dermatologists a second opinion based on the photos uploaded by patients, helping them spot things that might otherwise be missed.

The system also leaves room for improvement in how data is secured. While the current implementation uses encryption and role-based access control, future versions of the system could explore the use of blockchain technology to provide an immutable audit trail of who accessed or modified patient records and when. This would add an extra layer of trust and accountability to the system.

There is also an opportunity to expand the system to other clinics and health centres in Uganda. The modular design of the system means that it can be adapted to different clinical settings without rebuilding it from scratch. A multi-clinic version of the system could allow patients to share their records seamlessly when they are referred from one facility to another, which is a common occurrence in Ugandan healthcare.

Finally, future development should explore deeper integration with Uganda's national Health Management Information System (HMIS) and the Uganda EMR, so that data collected at Kampala Skin Clinic can contribute to national-level health reporting and planning in a standardized way.

REFERENCES / BIBLIOGRAPHY

APPENDIX

Appendix 1. Questionnaire
Appendix 2. Screenshots of sample code